

RESEARCH ARTICLE

A Meta-Adaptive Framework Combining Gated Attention Mechanisms and Feature-Wise Modulation for Multi-Horizon Stock Price Movement Prediction

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ABSTRACT Stock trend prediction aims to classify trading decisions into buy, hold, and sell signals with the goal of ensuring traders maximize profits. With the application of deep learning models and their efficiency across multiple tasks, there is an increasing research direction into their applications in stock trend prediction. However, most of these models are not trained to understand complex market structures and are sometimes oversimplified, making them not very useful in real-world trading performance. This study proposes a meta-adaptive multi-horizon forecasting model that integrates historical price data with domain-specific structural features to improve market trend prediction across 1, 3, 7, and 10-day horizons. The model utilizes fair value gaps (FVG) derived via a deterministic three-bar imbalance rule that captures liquidity voids, inefficiency zones, and institutional trading points. These features are also augmented by order block attributes and rolling-window hurst exponents that capture fractal persistence and market transitions in a principled form. The proposed model employs a variable selection network for dynamic soft-gating over the input set, and a multi-scale projection module that transforms each sequence using learned pooled-up sampling before fusing them through trainable attention weights. This fused representation is then passed through the feature-wise linear modulation block, which is adjusted based on a global context vector that enables horizon-specific adaptability. The temporal dependencies are modeled with a gated attention unit encoder and a gated residual decoder that classifies the buy, hold, and sell labels across different future horizons. The model was evaluated on eight indices, namely BSE, DJIA, FTSE, NASDAQ, NIFTY, N225, S&P500, and SSE, using time series cross-validation with $k = 3, 5$, and 7 fold settings. The results clearly indicate that the proposed model outperforms existing benchmark models in terms of average accuracy, precision, recall, and matthew correlation coefficient over all evaluated indices, with values of 94.97%, 95.33%, 95.07%, and 0.9250, respectively. A real-time trading simulation assesses the practical impact of the window size, considering 5, 10, and 15 days, with subsequent statistical validation through a paired t-test and Wilcoxon signed-rank test against benchmarks. Conclusively, this study provides valuable insights and demonstrates how both short and long-term traders can effectively leverage the proposed model using historical price data, market cues, and smart money concepts to accurately classify stock price movement.

INDEX TERMS Deep learning, fair value gaps (FVG), feature-wise linear modulation (FiLM), gated attention unit (GAU), stock price movement, order block.

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I. INTRODUCTION

Stock market prediction is one of the major tasks in recent financial engineering due to its applicability in the design of

trading strategies for prediction of price movements. These methods aim to give better returns on investment, taking advantage of market trends, optimizing portfolios, and avoiding unwanted risks of losses in the market [1], [2]. The importance of an effective and accurate model for trend prediction ensures that the trading decisions made by investors give maximized profits categorized into buy, hold, and sell labels [3]. However, it is often difficult to accurately predict the market because it is non-stationary and exhibits long-range dependency. Most financial markets are also influenced by the sentiments of different traders and participants, which are expressed in the form of financial news, company announcements, macroeconomic reports, or unexpected geopolitical events that contribute to price volatility [4]. As seen in many stocks, sentimental elements were consistently demonstrated during the COVID-19 market crash in 2020 where market sentiment affected both the buyers and sellers' decisions, leading to fluctuations in prices [5], [6]. All of these issues mean that institutional traders will need to employ predictive models to make high-frequency trades while retail traders will need to use predictive models to make swing trading or long-term investments. This means that these models need to examine a ton of relevant market data to learn and adapt to the market movements [7], [8].

The field has seen tremendous growth, all starting from traditional methods that applied simple econometrics models such as autoregressive conditional heteroscedasticity (ARCH), generalized ARCH (GARCH), and autoregressive integrated moving average (ARIMA) to capture market volatility and simple autocorrelation [9], [10]. These models typically assume linear relationships, stationarity, and capture little volatility clustering, which makes them ineffective, as these stationary conditions do not hold in real-world trading [11]. Various studies have consistently shown that stock returns typically exhibit asymmetric volatility, non-linear dependencies, and heavy tail patterns that lead to poor out-of-sample performance. As a result, machine learning methods such as support vector machines (SVM), random forests, and gradient boosting (e.g., XGBoost) were introduced to overcome these limitations [12], [13]. ML models showed improved performance in capturing these non-linear relationships, but the inherent complexity of the market microstructure limited the ability to model various market indicators, such as those derived from sentiment analysis or visual patterns [14].

More recently, there have been an increasing number of studies that have utilized deep learning models to capture complex patterns and enhance predictive capabilities. Among such models are long short-term memory networks (LSTM), convolutional neural networks (CNN), and transformers that are designed to capture long-range dependencies in stocks or spatial hierarchies or inputs [15], [16]. These models have been employed to model sentiments from news headlines and candle patterns through image recognition techniques. There have also been attempts in multi-modal fusion processing to merge historical data, text, and images.

Despite the extensive progress achieved using deep learning models, most of the research in the area either treat price as a single generic variable or not optimized for real-time trading utilizing market cues. This problem formulation can also often reduce the complexity of market dynamics without introducing structural information with available information useful to professional traders and quantitative analysts. The structural information includes features like fair value gaps (FVG), order blocks (OB), and liquidity zones that, because of the deep theoretical understanding of price efficiency and institutional activity, contain relevant information on market structure. FVGs are defined as a three-candle differential with a deterministic zone where the price is traded inefficiently and therefore there is high probability of a price reversal or mitigation [17]. The OB is for institutional accumulation and distribution with a large supply/demand imbalance [18]. Further, since market trends are divided by trending, ranging and mean-reverting, modeling the market would require structural cues which characterize each trend by considering fractal persistence in the market data. This would be better than strictly statistical methods that often miss such dynamics. Many stock trend prediction models are based on a static training workload and stationary deployment. In reality, the markets are evolving continuously and hence a fast adaptive model is necessary to remain robust in constantly changing markets. Meta-learning, particularly task-based learning, has recently become a prominent research direction in learning initialization parameters that have robust generalizations to new environments [19].

To overcome these limitations, this paper proposes a meta-adaptive multi-horizon forecasting architecture that integrates historical price, market structure cues, and meta-learning. The findings of this study can be summarized in the following manner:

- Inclusion of market structure cues like FVG and OB attributes that encode liquidity imbalances, institutional order flow, and specific market inefficiency zones. These are augmented by rolling-window hurst exponents that provide strong information on how the market is changing over time across different timeframes.
- Dynamic feature gating using a variable selection network that facilitates hierarchical temporal reasoning via trainable softmax weights by efficiently capturing short-term volatility and long-range trend dependencies simultaneously.
- Employment of gated attention units (GAU) to provide feature selectivity with linear complexity and feature wise linear modulation (FiLM) to incorporate global context information into the network through adaptive scaling and shifting. This provides good temporal modeling without the quadratic cost associated with the standard attention mechanisms.
- Integration of the almost-no-inner-loop (ANIL) inspired meta-learning strategy to address the non-stationarity in stock data. This enables the model to quickly adapt

to new market trends and significantly improve model generalization.

- Evaluation of the proposed model on eight indices using different time series cross-validation folds. The model performance is assessed using classification metrics (accuracy, precision, recall, f1-score, matthew correlation coefficient) and economic profitability (sharpe ratio, maximum drawdown, return on investment) over benchmarks.

The structure of the study is as follows. Section I gives a brief overview of the study, and Section II reviews related studies on stock trend prediction and meta-learning. Section III outlines the proposed methodology, which includes feature engineering and model architecture. Section IV describes the experimental setup and data preparation. Section V discusses the results and discussion, and Section VI concludes the study with suggestions for future work.

II. LITERATURE REVIEW

Various studies have explored the application of deep learning models for stock trend prediction due to their capability to handle and process non-linear patterns more effectively than statistical models [20]. However, these studies were generally single models based on historical price information and technical indicators and then switching to hybrid models so as to maximize efficiency of these single models [21].

For instance, the work by Shaban et al. [22] proposed a two-stage hybrid model using LSTM and bidirectional GRU. In this work, they focused on two main modules, where the first module deals with the data preprocessing stage involving feature selection, normalization, and cleaning. The second module is responsible for the prediction stage, which outperformed the benchmarks considered. Another hybrid model proposed by Ji et al. [23] is the galformer model, which enhances the transformer model with a generative one-step decoding approach to enable parallel multi-step predictions. They evaluated the model's performance on different indices (CSI300, DJIA, IXIC, S&P500), showing it outperformed various benchmarks like ARIMA, LSTM, transformer, and informer. Gezici and Sefer [24] generated image patches using the Swin Transformer model to convert historical prices. These patches were then passed through the ConvMixer, which is a patch embedding-based convolutional architecture that leverages inherent image-like properties. The model was evaluated using different exchange-traded funds categorized into buy, hold, and sell labels. The proposed model outperformed various benchmarks, including LSTM, vision transformers, data-efficient image transformers, and ConvMixer across both classification and profitability metrics. Hang et al. [25] proposed a multilevel graph attention network that extracts inter-stock relationships from knowledge graphs instead of correlation-based edges. The model constructs an isomorphic stock relation graph, capturing the different relation types among different stock types. The node

representations are derived from historical price data, and the model assigns different weights to each relation type, allowing the selective aggregation of relevant neighbor information. It was evaluated with a binary up/down classification and outperformed all benchmarks by over 12% accuracy. Wang [26] proposed hybrid gated linear unit architecture and a dual-stage temporal attention mechanism to predict cleaning resource stocks over a 16-day horizon. The model processes the input set by combining convolution operations with a dynamic gating mechanism and passes it through temporal attention via encoder hidden states. The decoder updates the states within an LSTM unit, which is then passed to the output layer. The model processes historical data across a 1 to 96 window interval and demonstrates superior performance compared to all the traditional methods. As already highlighted, most of the deep learning models focus more on technical indicators, typically ignoring market structure cues like OB and FVG that originate from trading strategies like smart money concepts.

Few studies have explored using different statistical methods that quantify market stability and predictability, such as hurst exponents, which show how price movement follows a trend, mean-reversion, or unpredictability at each time-step. A study by Cho and Lee [27] conducted a comparative analysis of different recurrent neural models (LSTM, BiLSTM, GRU, BiGRU) for predicting one-day ahead absolute returns of the S&P500. The models leveraged asymmetric hurst exponents derived using asymmetric multifractal detrended fluctuation analysis to capture long-range dependencies in either bullish or bearish markets. The proposed model was evaluated using different input variables: past returns, absolute returns, historical volatility, hurst exponent, and asymmetric hurst exponent. Another similar study using the hurst exponent is by Florindo et al. [28], who used an attention-based hurst exponent model for predicting the closing price of seven indices (DJIA, FTSE100, N225, SSE, S&P500, NASDAQ, and NYSE). Their model utilized a CNN layer for local pattern extraction from raw returns, multi-head attention to weight temporal dependencies, and a generalized hurst exponent for fractal input integration. The proposed model was shown to outperform all benchmark models. Both studies move beyond the typical simple one-day ahead prediction using only historical price data and technical indicators by building models that also examine other market trends to make more informed decisions.

The concept of meta-learning and how it functions and is applied are also discussed in this study, given the nature of the market and the effects of changing historical data on trading decisions. This challenge was addressed in a paper by Noor and Fatima [29] which proposed a model based on model-agnostic meta-learning, which can quickly adapt to changes and manage them, particularly considering the use of cryptocurrency for their study. They emphasize the importance of models being able to adapt to constantly changing market trends. Also, the study highlighted that future research could explore simulating similar models on trading platforms

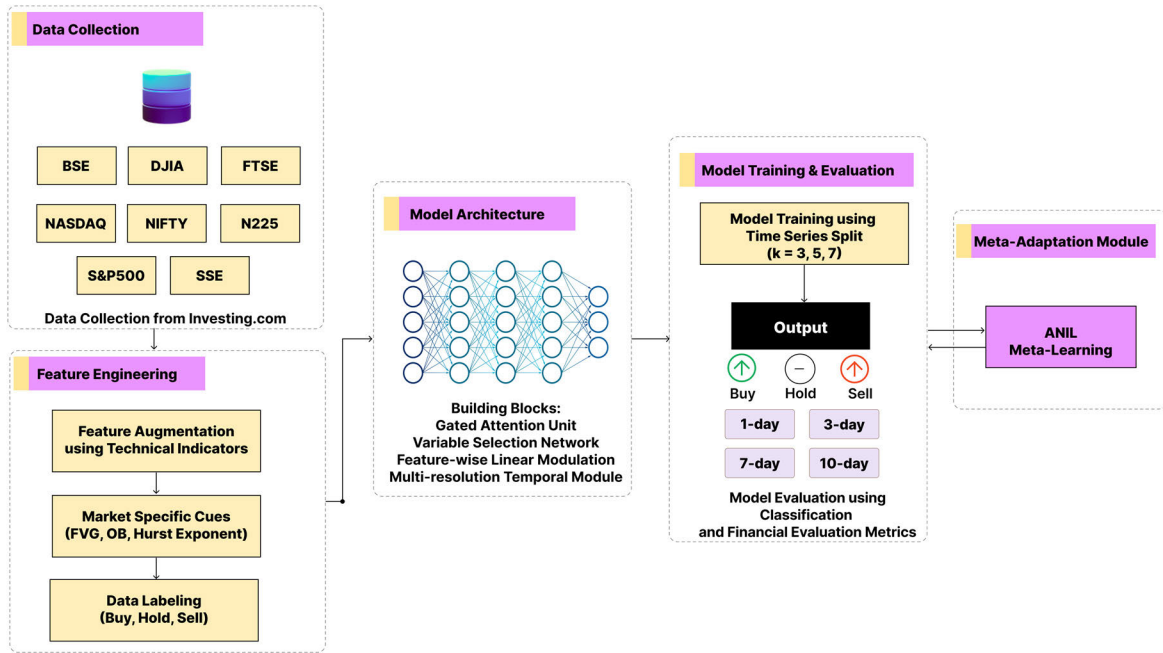


FIGURE 1. Overall proposed framework used in this study.

to see how the model adapts in entry and exit signals to optimize maximum profit. Another study by Liu et al. [30] proposed a meta-learning model that treats recent rolling windows as separate meta-tasks. This allows the model to learn an initialization from which an LSTM module adapts in only a few gradient steps to the latest price data. They also applied a variational mode decomposition to eliminate look-ahead bias when predicting next-day prices. This study proposes extending their work by replacing the LSTM base learner with a transformer-based module featuring gated attention and feature-wise modulation for better handling of long-range dependency in the data, considering the objective of predicting multi-horizon price movements.

III. METHODOLOGY

The methodology is designed to model temporal dependencies, structural market cues, and adaptive learning dynamics for multi-horizon stock price movement. The architecture efficiently handles historical OHLCV data, engineered market-structure cues, multi-scale temporal projection, dynamic feature gating, gated-attention sequence modeling, and a meta-learning adapter that improve market generalization. The architecture is optimized for multi-horizon classification tasks and integrated into a meta-learning framework to efficiently adapt to new tasks. This overall pipeline used in the study is depicted in **Figure 1**.

A. ROLLING-WINDOW HURST EXPONENT ESTIMATION

The hurst exponent is a statistical measure that quantifies the long-term pattern of a time series, indicating whether

the series is trending, mean-reverting, or random [31]. It is calculated using a rolling-window approach over a fixed-size window of recent data. Given a rolling window $W_t = \{P_{t-T+1}, \dots, \}$, the estimation uses the rescaled range (R/S) analysis as in **Equation (1) to (4)**:

$$Y_\tau = W_\tau - \bar{W} \quad (1)$$

$$Z_\tau = \sum_{i=1}^{\tau} Y_i \quad (2)$$

$$R = \max(Z_\tau) - \min(Z_\tau) \quad (3)$$

$$\log(R/S) = H \log T + C \quad (4)$$

The rolling window allows traders to see if the market is shifting from a trending ($H > 0.5$), favoring a buy signal, to a mean-reverting ($H < 0.5$), suggesting a sell signal. For risk management, taking new trading positions can be reevaluated if the H value is constantly changing, as this signals high uncertainty and instability in the market. The primary use of the H value is to act as a filter for other indicators and as an input feature to improve stock market predictability. Although it can be sensitive to window size, it is most effective when combined with other inputs rather than being the sole basis for trading decisions.

B. FAIR VALUE GAPS (FVG)

The FVG is a central concept within the smart money concept in trading, also known as a price imbalance or inefficiency [32]. It represents a price area where trading activity happens heavily in one direction, leaving a significant gap in liquidity. It is identified by a specific three-candle formation, as in **Figure 2**, where the strong middle candle (candle 2)

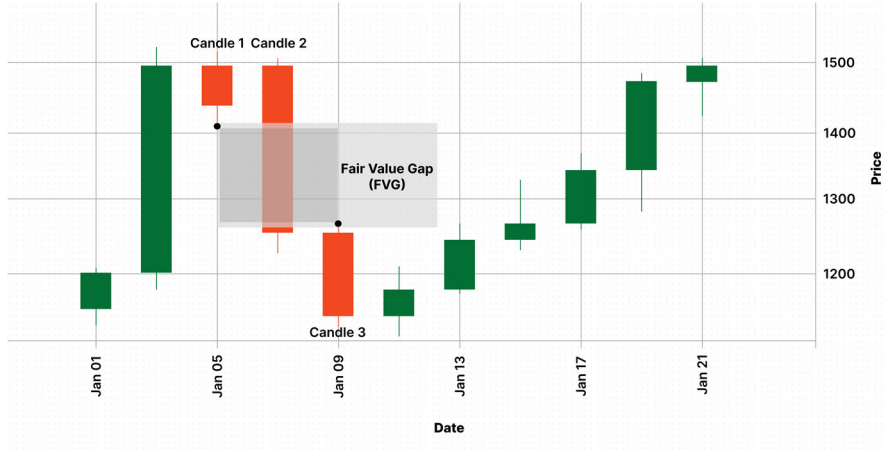


FIGURE 2. Price action showing fair value gap / price imbalance.

moves rapidly, typically indicating a large institutional order entry. The FVG is the vertical space between the low of candle 1 and the high of candle 3 for a

bearish FVG (or vice versa for a bullish FVG). This liquidity area is considered a market inefficiency because the price moved quickly without sufficient opposing pressure to create overlapping wicks between the candles. FVGs are interpreted as points of interest because the price is highly likely to return to and fill this position in the future. Institutions often use these gaps to take profits or rebalance their positions, expecting the price to return to the FVG to find support or resistance levels [33]. The deterministic three-candle imbalance is calculated as in **Equations (5) to (7)**, where it tests for either bullish or bearish FVG conditions. It computes the distance to the zone center by the number of active gaps and time since formation.

$$Low_{t-2} > High_t - \text{Bullish condition}$$

$$High_{t-2} < Low_t - \text{Bearish condition} \quad (5)$$

$$d_t^{FVG} = \frac{|P_t - c|}{P_t} \quad (6)$$

$$c = \frac{top + bottom}{2} \quad (7)$$

C. ORDER BLOCK (OB)

Similar to the FVG, an OB is a candlestick pattern that represents the price level where large institutional traders place significant clusters of buy or sell orders [34]. It is typically the last opposite-colored candle that occurs immediately before a strong and decisive move that breaks the current market structure. It is often referred to as a liquidity pool, which consists of a zone where orders have been initiated but not fully executed.

They are used and considered the origin of a large market move, serving as crucial support or resistance zones when the price revisits that area. This means OBs are only considered valid if the move originating from them successfully causes a break of structure that confirms a change in market direction.

An OB is identified when future returns exceed an average true range threshold as given in **Equation (8)**:

$$\left| \frac{P_{t+k} - P_t}{P_t} \right| > \lambda \cdot ATR_t \quad (8)$$

D. GATED ATTENTION UNIT

The gated attention unit (GAU) proposed by Taylor et al. [35] is one of the fundamental sequence processing blocks which is considered an efficient alternative to the traditional multi-head attention mechanism. It uses a gating mechanism with sigmoid gates for improved feature selectivity, which is fully parallelizable in training. It is useful for capturing cross-temporal interactions without quadratic complexity. Given an input $X \in \mathbb{R}^{T \times d}$, the GAU computes an expanded hidden representation as in **Equation (9)**, where $d_e = \alpha d$ is the expansion dimension and $\emptyset(\cdot)$ is $ReLU$.

$$H = \emptyset(XW_e + b_e) \quad (9)$$

A gating vector is then computed as **Equation (10)**:

$$\tilde{H} = HW_p + b_p \quad (10)$$

The GAU output is the gated mixture as in **Equation (11)**:

$$Z = G \odot \tilde{H} \quad (11)$$

Finally, a dropout and residual normalization is applied as in **Equation (12)**:

$$GAU(X) = LayerNorm(X + Dropout(Z)) \quad (12)$$

E. VARIABLE SELECTION NETWORK (VSN)

The variable selection network (VSN) is a learnable mechanism for dynamic feature weighting and selection, introduced in the temporal fusion transformer by Lim et al. [36]. It is the first adaptive layer that performs context-aware gating over the input set, which automatically learns to up-weight features that are informative and suppress those that add noise.

Given a multivariate time-series input at time t with J features as in **Equation (13)**:

$$x_t = [x_t^{(1)}, x_t^{(2)}, \dots, x_t^{(J)}] \in \mathbb{R}^J \quad (13)$$

The VSN computes a context-dependent feature importance vector, where v is a learned projection vector, GRU_{sel} is a small recurrent unit that processes the raw inputs and previous hidden state to extract contextual information as in **Equation (14)**.

$$w_t = \text{softmax} \left(v^T GRU_{sel} \left(x_t, h_{t-1}^{sel} \right) \right) \in \mathbb{R}^J \quad (14)$$

Each individual feature is then separately transformed by a specific sub-network, and the final selected representation is a soft-weight combination as in **Equations (15) to (17)**.

$$\tilde{x}_t^{(j)} = \text{Linear}_j \left(x_t^{(j)} \right) \quad (15)$$

$$z_t = \sum_{j=1}^J w_t^{(j)} \tilde{x}_t^{(j)} \quad (16)$$

$$z_t = \text{VSN} (x_t) = \tilde{X}_t w_t \quad (17)$$

F. PROPOSED MODEL ARCHITECTURE

The proposed architecture is designed to handle three main challenges, including multi-scale volatility, dynamic feature relevance, and long-term dependency modeling in the stock data at a minimal computational cost. The stock dataset contains historical data, technical indicators, and market structure cues that are often correlated or redundant. The VSN is used as a dynamic feature filter that prioritizes signal over noise by using a *softmax* gating vector at each time step.

Firstly, it summarizes the entire input sequence $X \in \mathbb{R}^{T \times F}$ (features across all T time steps) into a single feature vector h using mean pooling as in **Equation (18)**. The global context α_t is used to compute the aggregate global context given in **Equation (19)**, and the gating mechanism is given in **Equation (20)**.

$$h = \text{meanpool}_t (X_t) \in \mathbb{R}^F \quad (18)$$

$$\alpha_t = \text{softmax} (W_s X_t + b_s) \in \mathbb{R}^{T \times 1} \quad (19)$$

$$\hat{X} = X \odot \alpha_t \quad (20)$$

The multi-resolution temporal projection is utilized to extract temporal patterns across different frequencies, where single-resolution models fail to capture market causality. For example, a 1-day move is nested in a 5-day swing, which is nested in a 30-day cycle. The input sequence is processed at three resolutions, including high (H), medium (M), and low (L) all described in **Equations (21) to (23)**, which are fused using learnable weights.

$$H = \varnothing(W_h \hat{X}) \quad (21)$$

$$M = \text{Up}_2(\varnothing(W_m \text{AvgPool}_2(\hat{X}))) \quad (22)$$

$$L = \text{Up}_4(\varnothing(W_m \text{AvgPool}_4(\hat{X}))) \quad (23)$$

The three parallel features are combined using a set of SoftMax-normalized learnable weights, as given in **Equations (24) and (25)**.

$$\alpha = \text{softmax}(a) \quad (24)$$

$$Z = \alpha_1 H + \alpha_2 M + \alpha_3 L \quad (25)$$

Because the predictive relationship between the features is not static, the FiLM is utilized for generating a context vector derived from the sequence to dynamically generate scaling (γ) and shifting (β) parameters that modify the latent features ($\tilde{Z}$) as in **Equation (26)**.

$$\tilde{Z} = Z \odot (1 + \gamma) + \beta \quad (26)$$

The GAU encoder block replaces the QK^T attention matrix calculation with a channel-wise gating mechanism, which serves as a primary sequential processing unit for low-cost and effective long-range dependency modeling. The expansion, gating, projection, and residual connection are explained using **Equations (27) to (30)**.

$$u = \varnothing(W_u \tilde{Z}) \quad (27)$$

$$g = \sigma(W_g \tilde{Z}) \quad (28)$$

$$e = W_p(u \odot g) \quad (29)$$

$$E = \text{Layer Norm}(e + \tilde{Z}) \quad (30)$$

This is passed to an encoder-decoder temporal stack, which separates the learning of the global context of the encoder. The classification decision (buy, hold, or sell) is influenced by the most recent d steps price action. The decoder handles the sequence E by slicing it to take only the last k time steps as in **Equation (31)**. The decoder GAU ensures the model captures reversal patterns that influence price movement classification as in **Equation (32)**.

$$D_0 = E[:, -k :, :] \quad (31)$$

$$D = \text{Layer Norm}(\text{GAU}(D_0) + D_0) \quad (32)$$

The GRN utilizes a gated skip connection to improve the output D from the decoder stack, ensuring that both linear and nonlinear transformations are preserved and contribute to the output as in **Equation (33) to (35)**.

$$r = W_2 \varnothing(W_1 D) \quad (33)$$

$$g = \sigma(W_g D) \quad (34)$$

$$G = \text{Layer Norm}(g \odot s + (1 - g) \odot D) \quad (35)$$

The final classification head uses the last time step of the GRN output. The model is designed to output different prediction horizons as in **Equations (36) and (37)**.

$$h = G[:, -1, :] \quad (36)$$

$$y_\tau = \text{softmax}(W_\tau h) \quad (37)$$

The meta-learner is used for implementing a model-agnostic learning approach that enables the model to adapt quickly and effectively to new and unseen market data by only fine-tuning a small, critical subset of parameters. In the meta-learning

stage, the support set used was drawn exclusively from the most recent 300 samples of the train split, ensuring that only historical information relative to the query period is used. During the large-scale training, the encoder and feature extraction layers learn a set of generalized parameters, Θ_{base} . The meta-learner is then presented with recent data (the support set) that represents the current market trend. It freezes the majority of the network and allows only the VSN and final output layers (classification heads) to be updated. This is critical to the model's efficiency, as it enables the model to make accurate predictions even during live trading when the market makes rapid movements.

IV. EXPERIMENTAL SETUP AND DATA PREPARATION

This section details the experimental setup, dataset description, data labeling, and evaluation metrics used to assess the proposed meta-adaptive gated attention-FiLM framework (MG-FiLMT). The training and evaluation of all considered models were run using the time series split to prevent data leakage and to ensure all data splits following a forward-only manner without access to future time-step results. All experiments conducted in the study were done on Google Colab with the hardware and software specifications detailed in **Table 1**.

TABLE 1. Hardware and software specifications used for the experiments.

Configuration	Specifications
RAM	12.67GB
OS	Windows 11
CPU	Intel® Xeon® CPU @ 2.30 GHz
Python version	Python 3.10.12
GPU	Tesla T4 NVIDIA-SMI 535.104.05
Disk Space	107.72GB

A. PROBLEM STATEMENT

The objective of this study is to predict the multi-horizon stock price movement classification by integrating temporal dynamics, structural market cues, and multi-resolution feature modulation. Given a historical sequence $X_t = \{x_{t-L+1}, \dots, x_t\}$ comprising of OHLCV variables, technical indicators, and engineered market structure cues (FVG, OB, and Hurst exponents), the task is to predict the market direction class label $y_{t+h} \in \{Buy, Hold, Sell\}$ for multiple prediction horizons, $h \in \{1, 3, 7, 10\}$. The major problems in this study are handling the non-stationarity of stock data and the class imbalance between buy, sell, and hold labels. Also, most studies using DL models have rarely utilized structural cues like FVG and OB. The experimental setup is designed to evaluate the model's performance and improve generalization as the model handles new data.

B. DATA COLLECTION

The experiments were conducted using eight indices, including BSE, DJIA, FTSE, NASDAQ, NIFTY, N225, S&P500,

and SSE. All the datasets were collected from Investing from January 1, 2014 to September 30, 2025 [37]. The historical data contained price information and volume, with the summary statistics of each index described in **Table 2**. The price features were augmented using technical indicators including simple moving average (SMA), exponential moving average (EMA), volatility index (VIX), moving average convergence divergence (MACD), commodity channel index (CCI), bollinger bands (BB), average true range (ATR), relative strength index (RSI), and TRIX [38].

TABLE 2. Summary statistics of the selected indices.

Index	Description	Count	Mean	Std. Dev
BSE	Bombay Stock Exchange	2912	43096.77	16914.80
DJIA	Dow Jones Industrial Average	2913	27495.93	8226.32
FTSE	Financial Times Stock Exchange 100	2926	7176.29	679.64
N225	Nikkei 225 Stock Average	2839	24737.36	7168.60
NASDAQ	NASDAQ-100 Index	2912	9830.63	4630.39
NIFTY 50	National Stock Exchange of India 50	2867	13510.89	5499.54
S&P500	Standard & Poor 500	2912	3357.94	1229.78
SSE	Shanghai Composite Index	2593	6079.75	1011.95

C. DATA LABELING

The buy, sell, and hold labels are categorized using a threshold-based rule that converts n -day returns into categorical signals. This is a common and well-established approach called the triple-barrier method by Lopez de Prado [39], which is the exact conceptual foundation of the labeling used. It computes the return for each horizon, as shown in **Equation(38)**, and assigns the labels if the return meets the threshold defined in **Equation (39)**.

$$Return_n = \frac{close_t - close_{t-n}}{close_{t-n}} \quad (38)$$

$$y_i = \begin{cases} Buy, & \text{if } Return_n > 0.005 \\ Sell, & \text{if } Return_n < -0.005 \\ Hold, & \text{otherwise} \end{cases} \quad (39)$$

D. EVALUATION METRICS

All the conducted experiments and ablation study were assessed on both classification and profitability metrics. Among the classification metrics used were accuracy, recall, precision, f1-score, AUC score and matthew correlation coefficient (MCC). These metrics were selected primarily for the purpose of achieving a fair performance comparison of the model based on the class imbalance of the trading

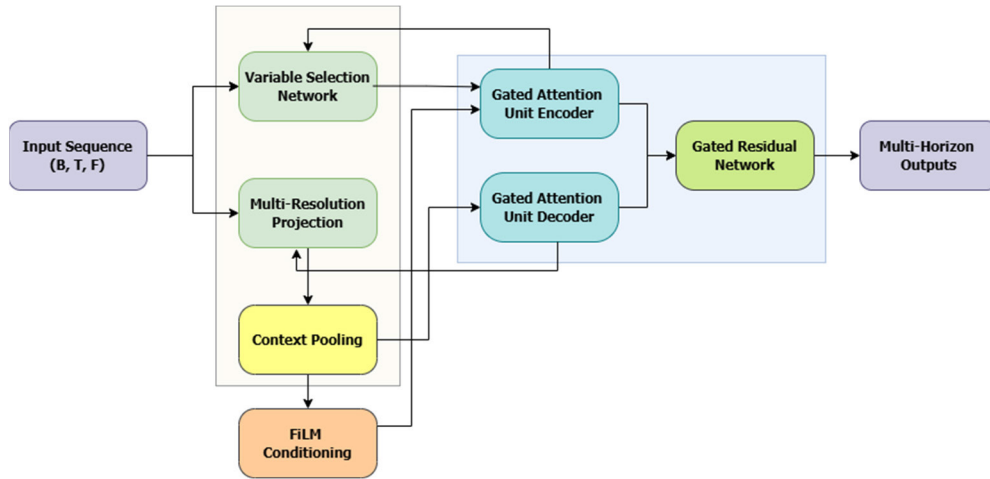


FIGURE 3. Proposed model architecture.

labels. A live trade simulation similar to the one used by Friday et al. [37] was used to examine the model in terms of profitability. Financial metrics including annualized return, return on investment (ROI), maximum drawdown (MDD), and the sharpe ratio (SR) were used to evaluate the model. Equations (40) to (48), as shown at the bottom of the next page.

V. RESULT AND DISCUSSION

This section describes the performance of the proposed MG-FiLMT model across different indices using a comparison across various CV folds (3, 5, 7) and the impact of window sizes (5, 10, 15). The time series CV was used to prevent data leakage, and the results are reported as averages over folds. These findings are then connected to the model's performance across multi-horizon predictions of 3, 5, and 10 days ahead, covering short-term market movement. It also benchmarks the model's performance against various state-of-the-art (SOTA) models and includes two statistical validation methods using the wilcoxon signed-rank and paired t-tests.

A. COMPARATIVE ANALYSIS ACROSS DIFFERENT CROSS-VALIDATION FOLDS

The first set of experiments was carried out to investigate the performance of the model across different CV fold splits, as in Table 3. For the BSE index, the accuracy steadily increases from 74.38% on the 3-fold CV, while the f1-score moves from 76.94% to 92.08%, and also MCC increases from 0.6771 to 0.8787. Each of these metrics was used to ensure that the results were not misleading, particularly considering the imbalance in price movement labels which might otherwise cause the model to favor the majority class. Given the BSE index has medium volatility and extended consolidation, the meta-learning process shows the adaptation to subtle changes in various bullish and bearish trends.

There was a similar strong performance on the DJIA, with an accuracy of 94.19% and an MCC of 0.9127 at the 7-fold CV. Similarly, all the indices showed improved classification results when moving from the 3-fold CV to the 7-fold CV. On the FTSE, the accuracy, f1-score, and MCC improved from 78.74% to 95.11%, 78.96% to 94.95%, and 0.6919 to 0.9255, respectively. On the NASDAQ, the accuracy and MCC improved from 76.07% to 93.35% and from 0.6938 to 0.9038. On the NIFTY, the accuracy and MCC improved from 75.51% to 87.94% and from 0.6827 to 0.8016. Lastly, on the SSE, the accuracy and MCC improved from 76.07% to 93.35% and from 0.7706 to 0.8694. Figure 4 shows the ROC curve for the proposed model using the 3-fold CV. Figures 5 and 6 show the confusion matrix and the ROC curve, respectively, for the 7-fold CV fold using a 10-day window size.

B. COMPARATIVE ANALYSIS USING DIFFERENT TIME STEPS ON 7-FOLD CV

Building on the results from the previous subsection, which identified the 7-fold CV as optimal for model performance, the next set of experiments evaluated performance across different window sizes of 5, 10, and 15 time steps. A consistent trend observed is larger windows outperform smaller ones consistent with the models being able to use longer historical sequences.

As for the BSE index, accuracy went from 88.89% with a 5 window to 92.06% with the 10-window, and finally to 94.31% with the 15-window size. The MCC also increased from 0.8456 between the 5 and 15 window sizes to 0.9135. The same improvements were also observed for the DJIA and FTSE, with high accuracies of 94.96% and 96.48%, respectively, on 15 window sizes. The model's accuracy with the 15-window size increased substantially from 85.29% to 97.05% on NASDAQ. Amongst the most obvious improvements were accurate results on the S&P500, from 74.57% to 95.28%. As demonstrated with all the indices in Table 4, the

model consistently showed better classification performance when the window size rose from 5 to 15. This suggests that longer sequences enable the model to learn better and predict better than smaller window sizes, while also allowing it to gain more historical context.

The results of this experiment can also be explained as data driven, since the volatility patterns that are simulated by the hurst exponent will differ greatly in prediction performance. Markets with higher persistence like FTSE and NASDAQ benefit from longer temporal windows, which allows the model to understand the direction and pattern of each trend effectively. As already emphasized, there is a clear difference in the classification performance with increasing window sizes, as smaller ones (5 days for instance) can be susceptible to noise that makes it difficult for the model to make any structural formation in the market.

C. CLASSIFICATION PERFORMANCE OF THE PROPOSED MODEL USING DIFFERENT WINDOW SIZES FOR 3, 7, AND 10-DAY AHEAD

The classification performance of the proposed MG-FiLMT model for multi-horizon prediction is evaluated for the 3, 7, and 10-day ahead horizons as reported in **Table 5**. It is observed that the performance generally declines as the prediction horizon increases. This is because market patterns are progressively affected by noise and structural changes in volatility, which, in turn, decreases classification performance. The longer window sizes of 10 and 15 steps are more effective compared to the shorter windows. Longer windows contain more relevant information, enabling the model to see more data to achieve better classification performance.

This was particularly evident in the indices like NASDAQ, FTSE, S&P500, and SSE, for which the 15-step window showed improved classification results in accuracy, recall,

and MCC, thus indicating that these markets have stronger medium-range temporal coherence and benefit from longer historical and contextual aggregation. For example, the FTSE is lower than 96.48% on the 3-day ahead horizon. It also falls to 93.38% on the 7-day ahead horizon and 89.66% on the 10-day ahead horizon using the 15-step window.

This is also true of NASDAQ where accuracy falls from 97.05% to 93.65% and 89.77% on the 3, 7 and 10 day ahead horizons. We observed that 5 time-steps efficiently capture immediate price action and volatility, the influence of which naturally reduces for longer prediction horizons. On the NASDAQ, S&P500, and SSE, the similarity in the results shows that irrespective of the different market trends, the model's performance is better for short horizons and slightly reduced at longer horizons.

D. PROFITABILITY ANALYSIS OF THE PROPOSED MODEL

This profitability evaluation uses a trading simulation using the classification results, as it establishes whether predictive improvements lead to economically meaningful trading outcomes.

The profitability analysis uses the real-time trading simulation algorithm from Friday et al. [38], starting with capital of ₹100,000, and incorporates a 0.1% transaction cost per trade as a proxy for the bid-ask spread. The trading strategy uses buy signals as entry trades, sell signals represent exit trades, and no signal represents a hold position. The multi-horizon predictors are also applied to inform position sizing where short-term signals trigger entry/exit, while longer predictors provide confirmation for holding periods. The position sizing is set at 10% of current equity, complemented by a 5% stop-loss with respect to MDD thresholds.

One consistent pattern across all the indices is that the shorter window of 5 steps provided higher absolute returns

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (40)$$

$$Precision = \frac{TP}{TP + FP} \quad (41)$$

$$Recall = \frac{TP}{TP + FN} \quad (42)$$

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (43)$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (44)$$

$$Annualized\ return = \frac{Final\ value}{Initial\ value}^{\frac{1}{n}} - 1 \quad (45)$$

$$ROI = \frac{Net\ profit}{Investment\ cost} \quad (46)$$

$$Sharpe\ ratio = \frac{Average\ return - Risk - freerate}{Std.\ Dev\ of\ returns} \quad (47)$$

$$MDD = \frac{Peak\ value - Troughvalue}{Peakvalue} \quad (48)$$

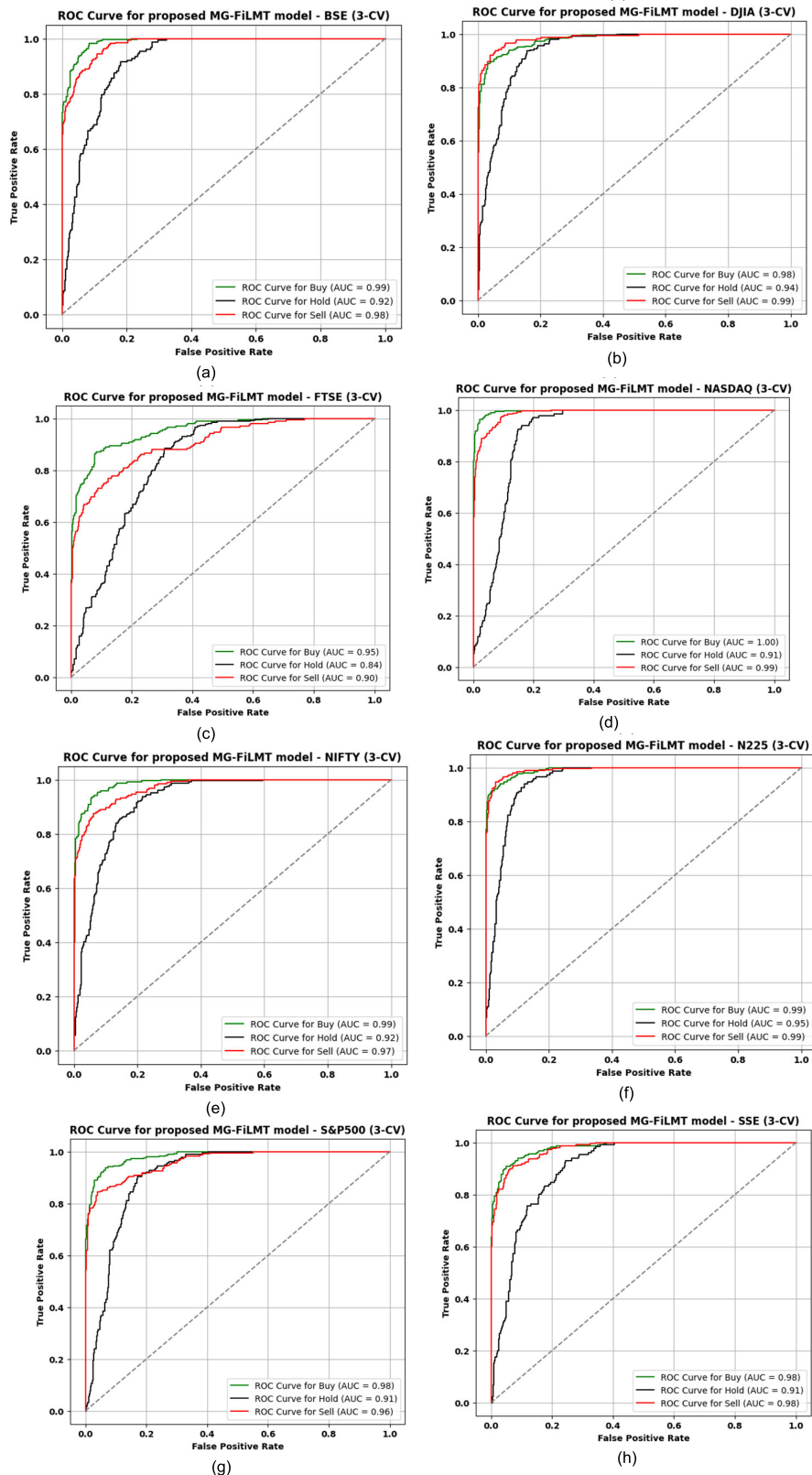


FIGURE 4. ROC Curve for the proposed MG-FiLMT model across 3-fold CV using 10 window sizes on (a) BSE (b) DJIA (c) FTSE (d) NASDAQ (e) NIFTY50 (f) N225 (g) S&P500 (h) SSE.

while the longer window (10–15 steps), show reduced draw-down and improve the sharpe ratio at the cost of lower ROI,

which reflects a basic trade-off in stock prediction. This is because short windows will adapt more aggressively to small

TABLE 3. Classification performance of the proposed MG-FiLMT model across different cross-validation.

	Accuracy	Precision	Recall	F1-score	AUC	MCC
BSE						
3-CV	74.78	88.14	80.14	76.94	96.98	0.6771
5-CV	80.26	89.28	80.35	81.86	97.03	0.7369
7-CV	92.06	92.11	90.98	92.08	98.66	0.8787
DJIA						
3-CV	84.27	88.33	88.96	84.49	96.89	0.7765
5-CV	88.26	88.22	89.15	87.63	97.93	0.8213
7-CV	94.19	94.15	95.20	94.13	99.49	0.9127
FTSE						
3-CV	78.74	82.86	79.80	78.96	93.44	0.6919
5-CV	86.48	88.09	87.51	86.66	97.86	0.8024
7-CV	95.11	95.22	96.53	94.95	99.75	0.9255
NASDAQ						
3-CV	76.07	89.35	76.84	78.19	96.48	0.6938
5-CV	82.07	89.65	82.33	83.19	96.55	0.7548
7-CV	93.35	94.50	92.70	93.47	98.54	0.9038
NIFTY-50						
3-CV	75.51	87.89	76.30	77.38	95.93	0.6827
5-CV	83.99	87.53	86.47	83.88	96.11	0.7733
7-CV	87.94	89.80	85.49	88.10	97.32	0.8016
SSE						
3-CV	85.11	85.71	84.89	85.36	95.46	0.7706
5-CV	90.73	90.83	91.57	90.77	96.51	0.8565
7-CV	91.50	92.03	91.95	91.68	97.98	0.8694

TABLE 4. Classification performance of the proposed MG-FiLMT model on 7-CV fold across different window sizes.

Window size	Accuracy	Precision	Recall	F1-score	AUC	MCC
BSE						
5	88.89	92.00	87.28	89.33	95.38	0.8456
10	92.06	92.11	90.98	92.08	98.66	0.8787
15	94.31	94.36	93.02	94.28	99.09	0.9135
DJIA						
5	90.21	91.25	88.53	90.45	98.12	0.8548
10	94.19	94.15	95.20	94.13	99.49	0.9127
15	94.96	95.05	95.72	94.97	99.60	0.9243
FTSE						
5	90.13	90.27	91.75	90.14	98.26	0.8525
10	95.11	95.22	96.53	94.95	99.75	0.9255
15	96.48	96.57	97.31	96.40	99.87	0.9467
NASDAQ						
5	85.29	90.26	86.19	86.31	96.53	0.7909
10	93.35	94.50	92.70	93.47	98.54	0.9038
15	97.05	97.18	96.84	97.08	99.56	0.9557
NIFTY-50						
5	76.05	75.64	78.09	75.77	92.02	0.6356
10	87.94	89.80	85.49	88.10	97.32	0.8016
15	90.72	92.31	91.28	90.93	98.48	0.8664
N225						
5	89.73	91.30	88.45	89.91	97.70	0.8516
10	91.39	91.41	90.83	91.39	98.81	0.8707
15	93.33	94.05	92.76	93.37	99.25	0.9027
S&P500						
5	74.57	87.66	77.71	76.41	94.35	0.6785
10	91.04	91.99	92.49	90.99	96.04	0.8643
15	95.28	95.43	96.10	95.33	98.55	0.9269
SSE						
5	94.67	94.78	95.26	94.50	98.69	0.9191
10	91.50	92.03	91.95	91.68	97.98	0.8694
15	97.66	97.65	97.50	97.64	99.84	0.9637

price fluctuations in an effort to capitalize on short-lived price differences, thereby increasing profitability by quickly reacting to sudden market swings.

However, this increased risk invariably leads to the model being very sensitive to high-frequency noise, increasing wrong market decisions. The longer windows incorporate

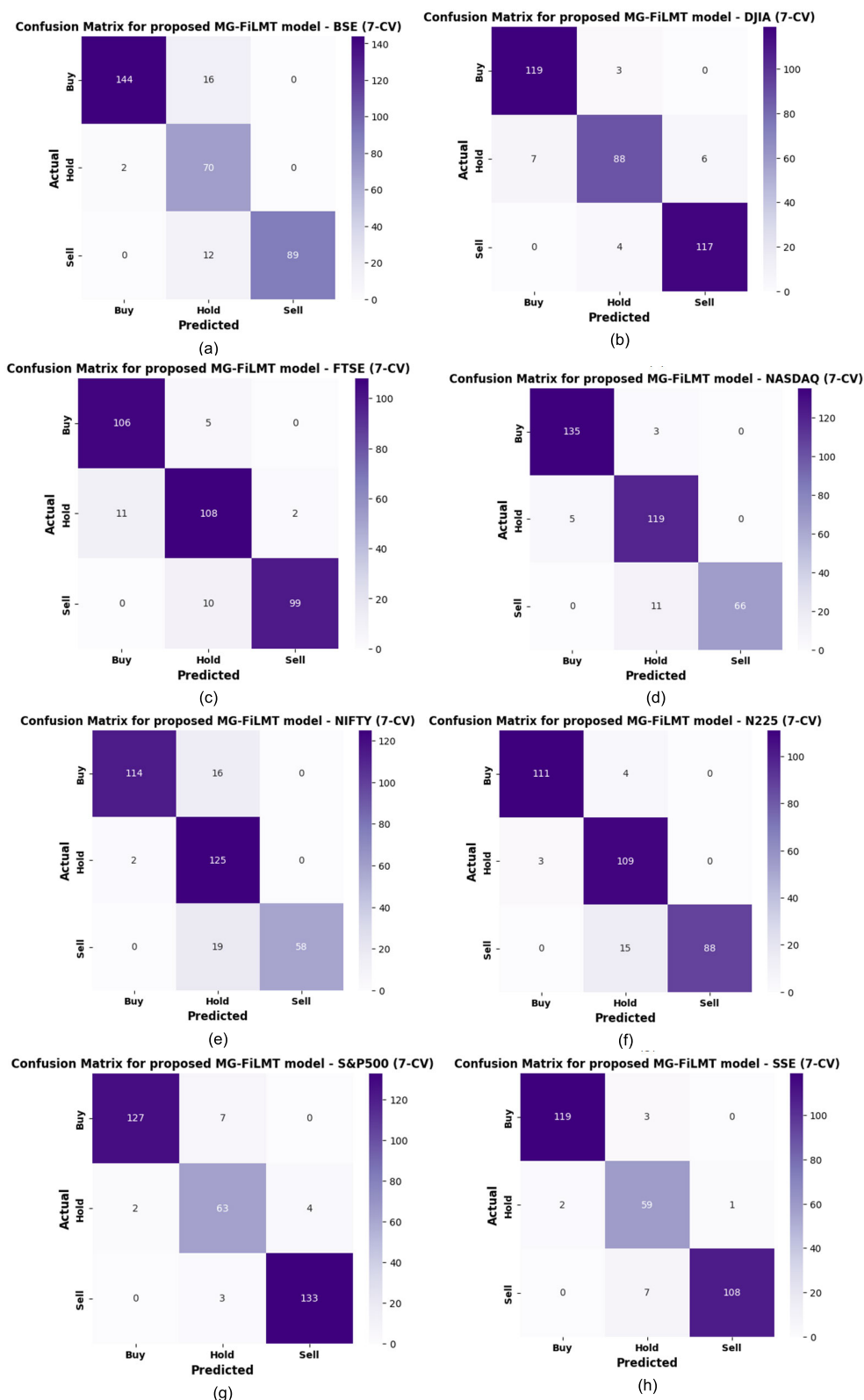


FIGURE 5. Confusion Matrix for proposed MG-FiLMT model across 7-CV fold using 10 window sizes on (a) BSE (b) DJIA (c) FTSE (d) NASDAQ (e) NIFTY50 (f) N225 (g) S&P500 (h) SSE.

more extensive historical data, smoothing short-term fluctuations and leading to better and more risk-controlled trading

strategy. For BSE, DJIA, FTSE, N225, S&P500, and SSE, the ROI for smaller windows is higher than larger window sizes,

TABLE 5. Classification performance of the proposed model using different window sizes for 3, 7, and 10-days ahead.

	3 days ahead				7 days ahead				10 days ahead			
	Accuracy	Precision	Recall	MCC	Accuracy	Precision	Recall	MCC	Accuracy	Precision	Recall	MCC
BSE												
5	88.89	92.00	87.28	0.8456	85.39	90.50	84.28	0.7856	81.19	89.00	81.88	0.6956
10	92.06	92.11	90.98	0.8787	88.76	90.61	88.08	0.8287	84.80	89.11	85.76	0.7537
15	94.31	94.36	93.02	0.9135	91.21	92.86	90.22	0.8735	87.49	91.36	87.98	0.8135
DJIA												
5	90.21	91.25	88.53	0.8548	86.71	89.75	85.53	0.7948	82.51	88.25	83.13	0.7048
10	94.19	94.15	95.20	0.9127	90.89	92.65	92.30	0.8627	86.93	91.15	89.98	0.7877
15	94.96	95.05	95.72	0.9243	91.86	93.55	92.92	0.8843	88.14	92.05	90.68	0.8243
FTSE												
5	90.13	90.27	91.75	0.8525	86.63	88.77	88.75	0.7925	82.43	87.27	86.35	0.7025
10	95.11	95.22	96.53	0.9255	91.81	93.72	93.63	0.8755	87.85	92.22	91.31	0.8005
15	96.48	96.57	97.31	0.9467	93.38	95.07	94.51	0.9067	89.66	93.57	92.27	0.8467
NASDAQ												
5	85.29	90.26	86.19	0.7909	81.79	88.76	85.19	0.7309	77.08	87.26	82.89	0.6409
10	93.55	94.50	92.70	0.9038	90.05	92.40	90.10	0.8538	85.13	89.00	86.26	0.7688
15	97.05	97.18	96.84	0.9557	93.65	95.38	94.04	0.8957	89.77	93.18	91.12	0.8217
NIFTY-50												
5	76.05	75.64	78.09	0.6356	72.55	74.14	76.09	0.5756	67.85	71.14	72.95	0.4856
10	87.94	89.80	85.49	0.8016	84.64	87.30	83.09	0.7416	79.32	83.60	78.07	0.6466
15	90.72	92.31	91.28	0.8664	87.22	89.81	88.18	0.8064	81.66	86.50	84.40	0.7064
N225												
5	89.73	91.30	88.45	0.8516	86.23	89.80	85.45	0.7916	82.03	88.30	83.05	0.7016
10	91.39	91.41	90.83	0.8709	88.09	89.91	87.93	0.8207	84.13	88.41	85.61	0.7457
15	93.33	94.05	92.76	0.9026	90.23	92.55	89.96	0.8627	86.51	91.05	87.72	0.8027
S&P500												
5	74.57	87.66	77.71	0.6785	71.08	86.16	74.72	0.6185	66.87	84.66	72.31	0.5285
10	91.04	91.99	92.48	0.8643	87.74	90.49	89.59	0.8143	83.78	88.97	87.26	0.7393
15	95.28	95.43	96.10	0.9269	92.15	93.93	93.30	0.8869	88.46	92.43	91.08	0.8269
SSE												
5	94.67	94.78	95.26	0.9191	91.14	93.28	92.26	0.8591	86.97	91.78	89.86	0.7691
10	91.50	92.03	91.95	0.8694	88.20	90.53	89.05	0.8194	84.24	89.03	86.73	0.7444
15	97.66	96.65	97.50	0.9637	94.56	96.15	94.70	0.9237	90.83	94.65	92.46	0.8637

as demonstrated with BSE values of 79.78% at 5 window size, compared to 20.12% at 15 window size. This suggests that small historical sequences can lead to more frequent and timely trades as they capitalize on a short-run volatility detected by the model. But it is the case that with larger windows a greater ROI on the NASDAQ and NIFTY-50, represent short-term trades with more false entries and exits. This is explained by NASDAQ increasing from 19.77% to 41.89% and NIFTY-50 rising from 68.83% to 75.78% where extended sequences best illustrate the persistence of trends across these highly volatile sectors. These profitability results in **Table 6** are implicitly connected to classification performances in

Tables 3 to 5. The aggregated improvements are directly related to better ROI and lower MDD, since the higher f1-scores and lower MCCs prevent false trades, which exhaust the trading capital.

The results of this analysis reflect the relationship between window size, price movements, and model performance, as well as its application to live trading conditions.

The model architecture with MRP layers showed good capture of relevant information across different horizons that allow for both long- and short-term price movements. **Figures 7 and 8** show the trading signals of the proposed model for 1-day and 10-day ahead predictions. For indices such

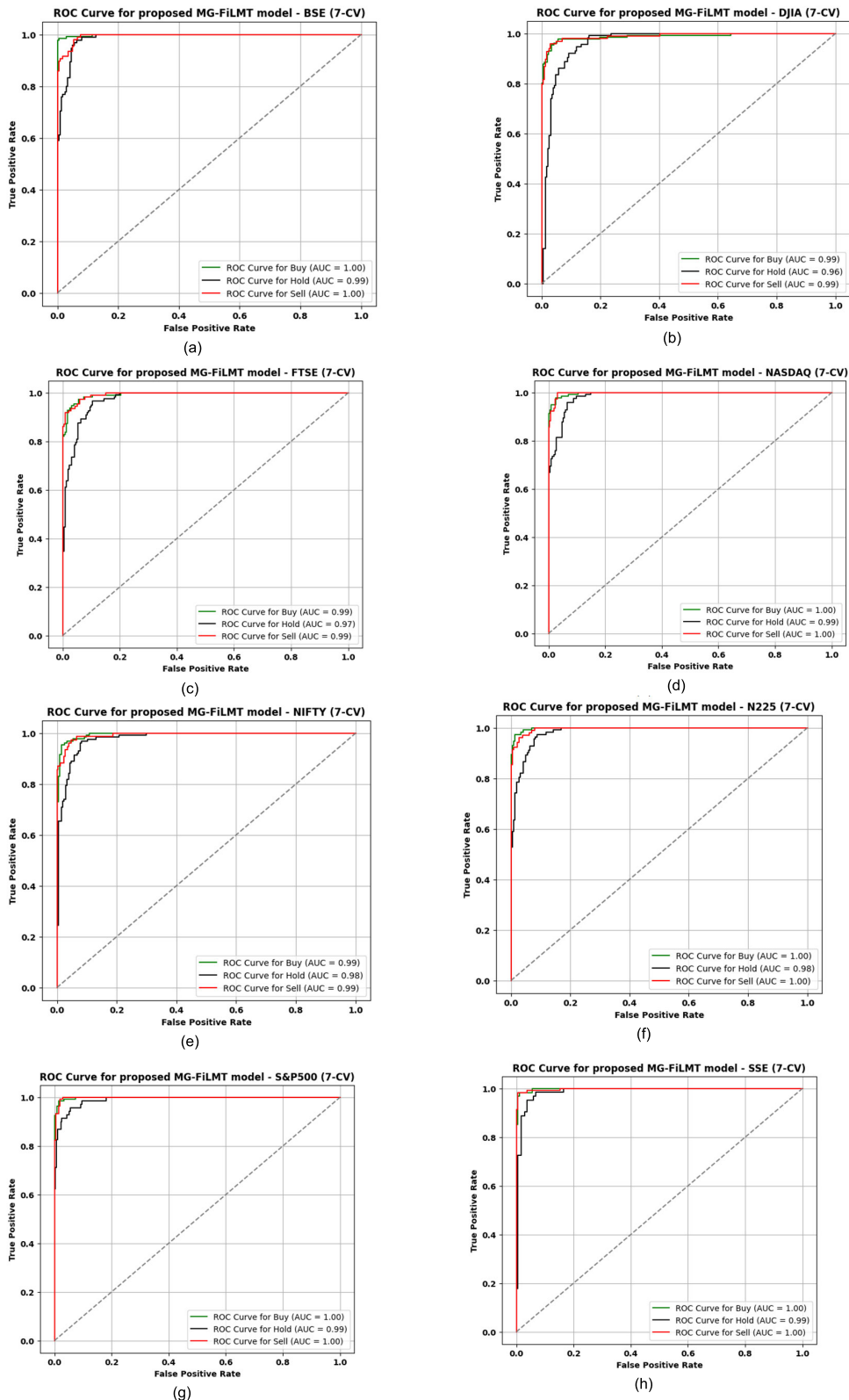


FIGURE 6. ROC curve for proposed MG-FiLMT model across 7-CV fold using 10 window size on (a) BSE (b) DJIA (c) FTSE (d) NASDAQ (e) NIFTY50 (f) N225 (g) S&P500 (h) SSE.

as the NASDAQ and NIFTY50, the shorter window sizes did not produce greater profits, even in price ranges, and

the model reacted slowly in market conditions. For example, MCC of NASDAQ at a window size of 15 is the highest

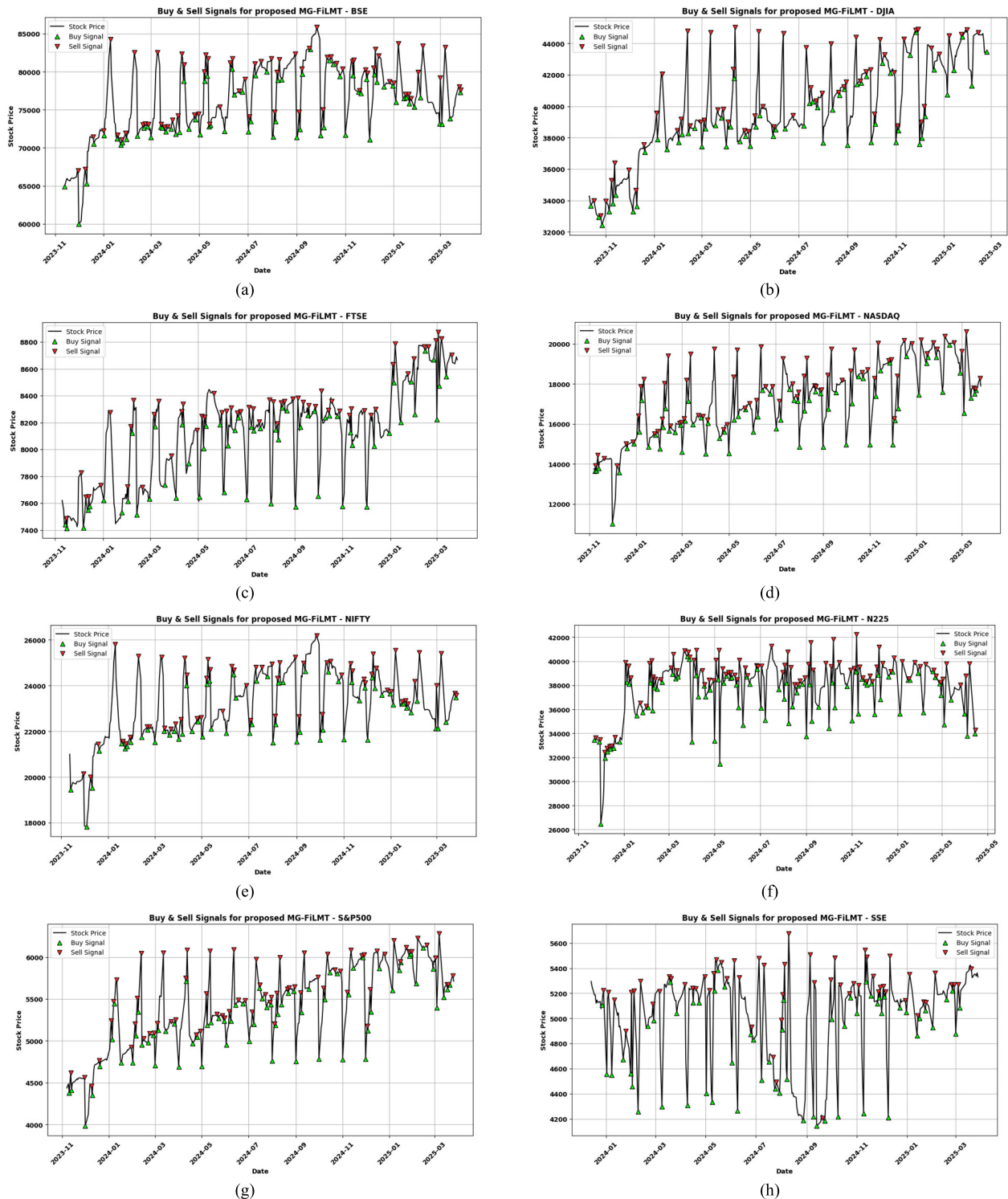


FIGURE 7. Trading buy and sell signals for proposed MG-FiLMT model across 7-CV fold using 15 window size for 1-day ahead on (a) BSE (b) DJIA (c) FTSE (d) NASDAQ (e) NIFTY50 (f) N225 (g) S&P500 (h) SSE.

long-horizon value (0.8217 for 10 days), equivalent to the highest sharpe ratio (1.9981). The BSE index did indeed predict the future accurately for all future periods, and this strong short-run performance resulted in high AR. The trading

simulation explicitly incorporates market structure cues, particularly the Hurst exponent and FVGs, which strongly influence the behavior of the proposed MG-FiLMT model in real-time decision-making. This adaptability reduces the

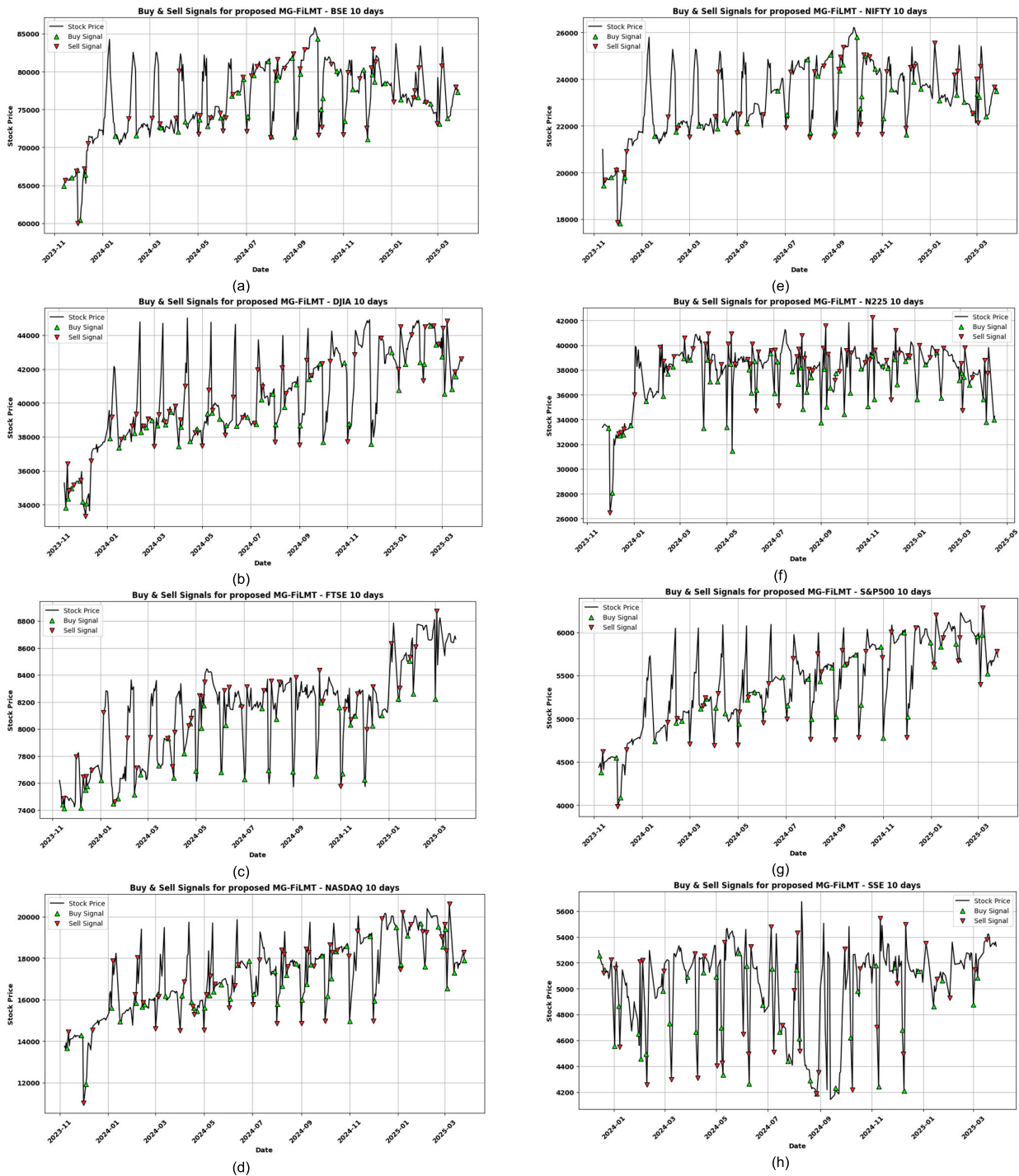


FIGURE 8. Trading buy and sell signals for proposed MG-FiLMT model across 7-CV fold using 15 window size for 1-day ahead on (a) BSE (b) DJIA (c) FTSE (d) NASDAQ (e) NIFTY50 (f) N225 (g) S&P500 (h) SSE.

likelihood of entering trades during noise-dominated phases or market reversals, thereby reducing unnecessary exposure.

E. ABLATION STUDY

The ablation study was conducted in order to quantify the contribution of each architectural component within the proposed MG-FiLMT framework. Three variants were

considered with identical hyperparameters as detailed in **Table 8** and evaluated for the same 7-fold CV using a 15-day window. **Table 7** summarizes the results across selected indices BSE, FTSE, NASDAQ, N225, and S&P500. FVG features tend to be weighted more strongly during periods of high volatility such as NASDAQ and N225, which shows how they capture short-term price inefficiencies

TABLE 6. Profitability analysis of the proposed MG-FiLMT model using different window sizes across all indices.

Window size	ROI	AR	MDD	Sharpe ratio
BSE				
5	79.78	45.05	21.95	1.6460
10	50.34	36.31	19.88	1.6153
15	20.12	15.98	18.11	1.6207
DJIA				
5	28.66	10.12	21.47	1.3121
10	18.77	9.41	19.33	1.4847
15	12.45	8.92	17.86	1.5728
FTSE				
5	35.06	11.50	23.42	1.5582
10	21.35	11.12	21.19	1.5807
15	16.99	10.34	19.75	1.8406
NASDAQ				
5	19.77	12.02	20.14	1.6110
10	27.45	12.88	22.10	1.7564
15	41.89	13.56	24.52	1.9481
NIFTY-50				
5	68.83	23.39	18.92	1.5863
10	71.14	24.42	20.21	1.5939
15	75.78	28.80	22.36	1.6337
N225				
5	30.48	10.44	21.31	1.5876
10	20.15	9.85	19.76	1.4521
15	13.92	9.21	18.45	1.3368
S&P500				
5	32.74	11.82	21.66	1.4183
10	21.56	10.21	19.44	1.5012
15	14.87	9.68	17.92	1.6769
SSE				
5	24.92	9.89	24.66	1.4266
10	15.77	8.21	22.34	1.1761
15	11.04	7.65	20.88	1.1028

and liquidity imbalances. For comparison, despite stable or trending market conditions, the hurst exponent has greater weighting as it effectively captures long-run dependence and persistence in price movement.

A similar observation is seen for the FTSE (92.30% to 96.48% accuracy, MCC from 0.8797 to 0.9467) and NASDAQ (91.42% to 97.05% accuracy, MCC from 0.8735 to 0.9557). The proposed MG-FiLMT consistently outperformed both ablated variants across all indices, confirming that the GAU indeed plays the central role in improving temporal feature interactions. This emphasizes the ability of the model in filtering noisy fluctuations that leads to improved classification performance in accuracy, AUC, and MCC, especially for the two volatile indices NASDAQ and SSE. Additionally, the VSN model was very useful for dynamic feature weighting within the proposed MG-FiLMT model. As illustrated in **Figure 9**, the VSN assigns feature weights across different CV folds and market indices in a consistent manner. We observe that features that are more important to the model's classification performance weigh more heavily, suggesting that feature relevance are stable rather than random. The hurst exponent and FVG structural feature also consistently have higher importance scores on all the indices, which demonstrates their potential to capture long-run

dependencies and market inefficiencies. But, the most obvious performance loss was when the multi-resolution components were removed, in the FiLM +TFT variant. This indicates that the multi-scale representations played a key role in recording short, medium, and long-range patterns which were necessary for better prediction. Each architectural block contributes independently and enhances the cumulative results across multiple indices.

Finally, in order to assess the meta-learning adaptation ability of the proposed model using a convergence analysis, as illustrated in **Figure 10**. The graph compares the proposed MG-FiLMT model to the baselines (FiLM + TFT, MRP + VSN). The results show that the proposed model converges better and reaches a stable accuracy plateau with fewer adaptation epochs compared to the baseline models. This demonstrates that the selective parameter update strategy enables stable adaptation to new markets without overfitting and validates the model's suitability in trading environments.

TABLE 7. Ablation study results for the proposed MG-FiLMT model.

Model	Accuracy	Precision	F1-score	MCC	Inference Time (s)
BSE					
FiLM + TFT	90.12	91.18	90.35	0.8422	0.0026
MRP + VSN	92.45	92.60	92.33	0.8750	0.0031
MG-FiLMT	94.31	94.36	94.28	0.9135	0.0044
FTSE					
FiLM + TFT	92.30	93.01	92.39	0.8797	0.0028
MRP + VSN	93.44	93.60	94.48	0.9061	0.0033
MG-FiLMT	96.48	96.57	96.40	0.9467	0.0046
NASDAQ					
FiLM + TFT	91.42	93.50	91.61	0.8735	0.0030
MRP + VSN	93.88	94.14	94.62	0.9221	0.0035
MG-FiLMT	97.05	97.18	97.08	0.9557	0.0049
N225					
FiLM + TFT	89.80	90.77	89.86	0.8324	0.0027
MRP + VSN	91.24	91.85	91.38	0.8718	0.0032
MG-FiLMT	93.33	94.05	93.37	0.9027	0.0046
S&P500					
FiLM + TFT	90.71	92.11	91.00	0.8461	0.0028
MRP + VSN	92.04	92.38	93.27	0.9044	0.0034
MG-FiLMT	95.28	95.43	95.33	0.9269	0.0048

F. COMPARISON WITH STATE-OF-THE-ART MODELS

The comparative study with SOTA models, summarized in **Table9**, highlights the advances that the proposed MG-FiLMT model offers for multi-horizon stock price movement classification. The SOTA models represent the latest developments from 2021 to 2024 and include convolutional-recurrent hybrids such as CNN-LSTM, ensemble methods represented by XGBoost, attention-enhanced architectures such as PLSTM-TAL, GA-TDDPL-CNN-LSTM and CAGTRADE. The earlier deep learning architectures, such as CNN, CNN-LSTM for the NIFTY50, and even hybrid ANN +IFS model, showed a wide variation in predictive performance, ranging from an accuracy of 53% - 91%. Among them, the best-performing previous model was PLSTM-TAL, which reported an f1-score of 86.76 but still

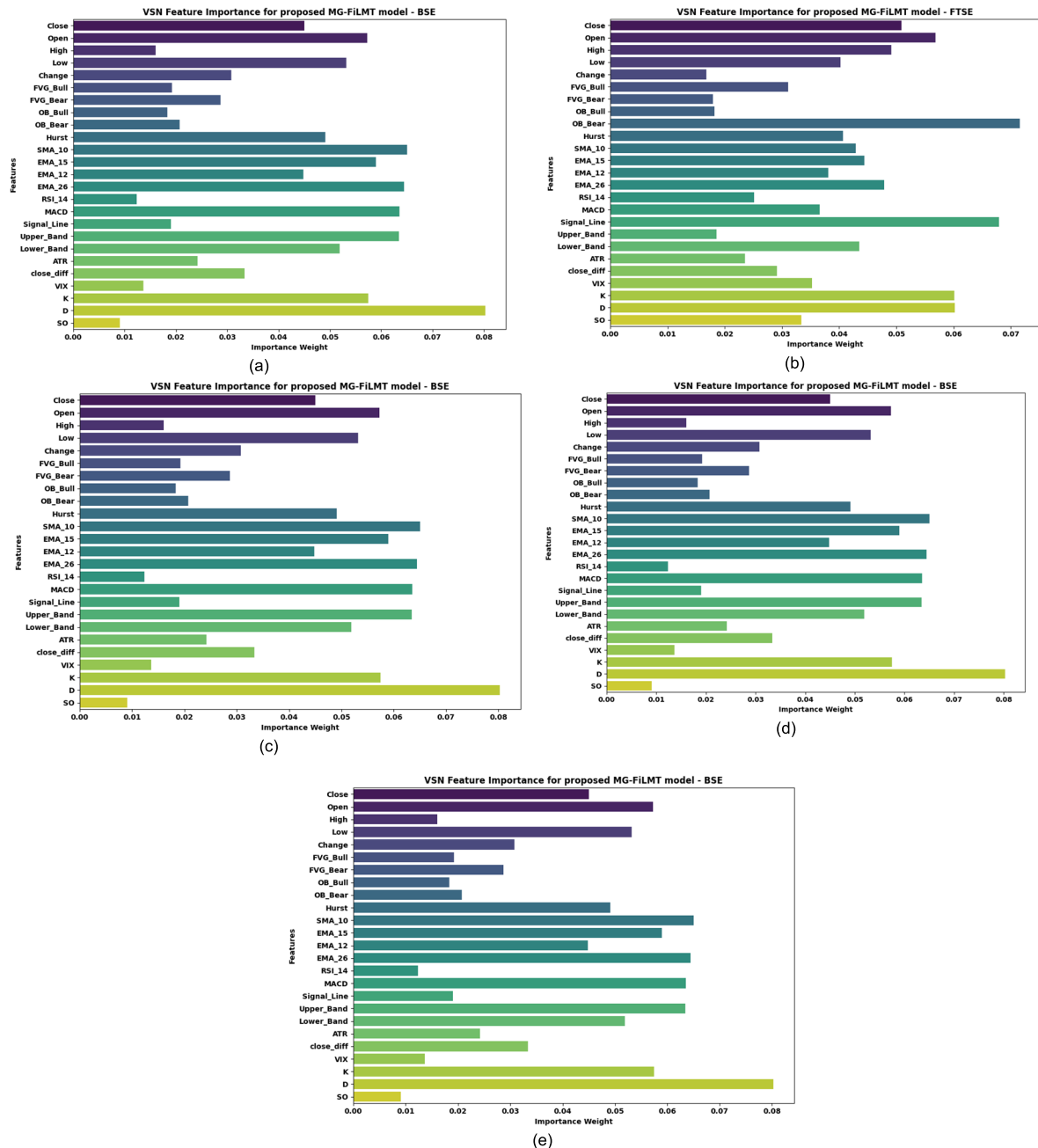


FIGURE 9. Dynamic feature importance by the Variable Selection Network across different indices (a) BSE (b) FTSE (c) NASDAQ (d) N225 (e) S&P500.

failed to maintain high discrimination power as reflected by its MCC of 0.6988. In contrast, the proposed MG-FiLMT model achieves an accuracy of 90.72%, while the balanced f1-score is 90.93, with significantly better separability of three movement classes reflected by an AUC of 98.48.

For the S&P500, the model achieves an accuracy of 95.28%, f1-scores of 95.33%, AUC of 98.55% and MCC of 0.9269, outperforming CAGTRADE (91.00%) and PLSTM-TAL (85.09% with MCC of 0.6975). These results indicate that the proposed MG-FiLMT consistently outperformed

SOTA by 4 - 40% in accuracy and up to 38% in MCC across different indices. While all three baseline models (TFT, PLSTM-TAL, and CAGTRADE) utilize attention mechanisms, they are trained in a traditional single-task setting. In contrast, the proposed MG-FiLMT model makes a positive contribution through adopting an ANIL-based meta-learning approach. This allows for transferable initialization that enables the model to easily adapt to new markets. Lastly, the VSN and FiLM layers are used to allow the model to dynamically adjust features relevance across multiple

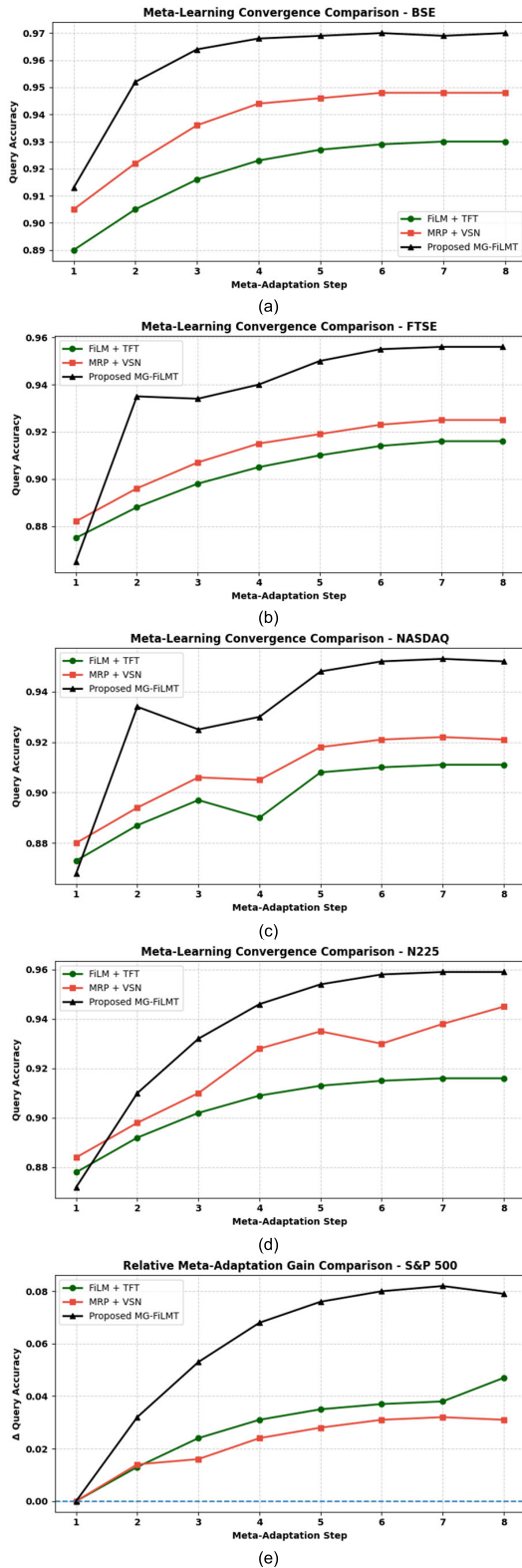


FIGURE 10. Meta-learning convergence across inner-loop adaptation steps across different indices (a) BSE (b) FTSE (c) NASDAQ (d) N225 (e) S&P500.

horizons, rather than relying on a single attention mechanism. The proposed architecture is not a new addition to the existing

work but is a meta-adaptive model designed to provide better generalization than the existing SOTA models.

G. STATISTICAL VALIDATION OF THE PROPOSED MODEL

Our method of statistical validation is to check the statistical significance of the performance improvement from the proposed model relative to benchmark studies. In the paired t-test, we compared the proposed model against two standard model versions: FiLM + TFT and MRP + VSN, based on performance scores for five indexes (BSE, FTSE, NASDAQ, N225, S&P500) [50].

The mean accuracy difference was 4.42 for the proposed MG-FiLMT model compared with FiLM +TFT with t -statistic 12.79 and p -value 0.0002, indicating significant superiority. The precision showed a mean difference of 3.40 ($t = 36.59$, $p < 0.0001$), f1-score 4.25 ($t = 12.81$, $p = 0.0002$), and MCC 0.074 ($t = 24.57$, $p < 0.0001$). These results are consistent with strong improvements when MRP, VSN, and GAU are taken as well as results with high t -values reflecting large effect sizes and low variable across the indexes as in **Table 10**. The comparison also showed that the proposed model against MRP + VSN had mean differences of 2.68 for accuracy ($t = 9.18$, $p = 0.0008$), 2.60 for precision ($t = 9.85$, $p = 0.0006$), 2.08 for f1-score ($t = 21.01$, $p < 0.0001$), and 0.033 for MCC ($t = 10.43$, $p = 0.0005$).

A second test, the wilcoxon signed-rank test, was also conducted to compare the proposed MG-FiLMT model to benchmarks [37]. It also outperformed both FiLM + TFT and MRP + VS variants with a p -value of 0.03125 and $W = 0$, indicating the results are statistically significant ($p < 0.05$). Both tests are rigorous enough to show that architectural improvements are not accidental. It is also notable that the combined effects of the models are positive and therefore strong over time, as the performance among different measures is consistent.

H. LIMITATION OF THE STUDY

Despite the promising performance of the proposed MG-FiLMT model, there are some limitations that must be noted to further contextualize the results and offer avenues of further research.

- 1) First, this study did not include slippage costs in the live trade simulation. Slippage is defined as the price difference between the expected execution price and the actual fill price. This is most common in highly volatile markets such as the NASDAQ and N225 where liquidity is fragmented and orders-book movements can be so fast that execution is inefficient.
- 2) Second, it is quite clear that the MG-FiLMT architecture is expensive to compute. In particular, the use of market structure cues, such as hurst exponent and FVG structure results in a large input space, which requires a substantial GPU memory and training time. Although this complexity is useful for improving predictive performance, it may prevent its deployment in

TABLE 8. Hyper-parameters of the proposed model and all compared models.

Hyper-parameters	Proposed MGFiLMT	FiLM + TFT	MRP + VSN
Decoder length	8	8	8
Model dimension	64	64	64
Dropout rate	0.1	0.1	0.1
GAU expansion factor	4	4	×
GAU layers	2 (1 encoder + 1 decoder)	2 (1 encoder + 1 decoder)	×
VSN hidden units	64	×	64
VSN projection dimension	64	×	64
MRP resolutions	High/ Medium/ Low	×	High/ Medium/ Low
Pooling factors	2×, 4×	×	2×, 4×
Trainable fusion weights	3-dim vector	×	3-dim vector
FiLM context hidden	Dense (64, ReLU)	Same	Same
Encoder	GAU (64 – 256 – 64)	GAU (Same)	Linear - Dense (64) - LayerNorm
Meta-learner LR	1e ⁻⁴	1e ⁻⁴	1e ⁻⁴
Meta-learner epochs	8	8	8
Optimizer	Adam	Adam	Adam
Approx. Parameters	293,571	167,865	203,453

TABLE 9. Performance metrics of the proposed model against state-of-the-art (SOTA) models.

Ref	Year	Model	Accuracy	F1-score	AUC	MCC
NIFTY50						
[40]	2022	CNN-LSTM	57.20	53.10	-	-
[41]	2022	Ensemble XGB	53.00	-	-	-
[42]	2023	CNN	78.00	-	-	-
[43]	2023	ANN + IFS	91.00	91.00	-	-
[44]	2024	PLSTM-TAL	85.11	86.76	91.95	0.6988
Proposed	2025	MG-FiLMT	90.72	90.93	98.48	0.8664
SSE						
[43]	2021	ANN + IFS	90.00	90.00	-	-
[45]	2022	Multiple Attention GNN	66.54	-	-	-
[46]	2023	CS-ACNN	57.30	-	56.80	-
[44]	2024	PLSTM-TAL	88.21	88.80	94.64	0.7637
Proposed	2025	MG-FiLMT	97.66	97.64	99.84	0.9637
S&P500						
[47]	2021	2D-CNN	75.38	65.25	62.16	-
[41]	2022	Ensemble XGB	54.00	-	-	-
[48]	2023	GA-TDDPL-CNN-LSTM	80.00	-	-	-
[49]	2024	YOLOv8 + VGG16	70.10	69.90	69.900	-
[44]	2024	PLSTM-TAL	85.09	86.71	93.12	0.6975
[38]	2024	CAGTRADE	91.00	91.50	-	-
Proposed	2025	MG-FiLMT	95.28	95.33	98.55	0.9269

low-resource environments or applications where ultra-low-latency inference is a critical feature.

- 3) The labeling scheme used in this study employs a fixed return threshold for buy, hold, and sell classes. This

method is consistent with all indexes and allows for a fair reflection of the model's meta-adaptive performance. It could be extended to an adaptation with different thresholds or a dynamic threshold based on

TABLE 10. Statistical significance of the proposed MG-FiLMT model.

Metric	Mean difference	t-stat	p-value
MG-FiLMT against FiLM-TFT			
Accuracy	4.42	12.79	0.00021
Precision	3.40	36.59	$3.33 \times 10e-6$
F1-score	4.25	12.81	0.00021
MCC	0.0743	24.57	$1.63 \times 10e-5$
MG-FiLMT against MRP + VSN			
Accuracy	2.68	9.18	0.00078
Precision	2.60	9.85	0.00059
F1-score	2.08	21.01	$3.03 \times 10e-6$
MCC	0.0322	10.43	0.00048

average true range, that would greatly affect the class distribution and decision boundaries.

- 4) Lastly, market microstructure noise such as limit order book imbalance, bid-ask spread dynamics or intra-day liquidity cycles is not explicitly modeled. These microstructural factors tend to influence the kinetics of the process in their later formative stages and if left out, the model might be less responsive to ultra-short-horizon signals. Although the addition of structural features from FVG captures some features of price inefficiency, the very absence of microstructure inputs may still limit the generalization ability of the model to high frequency trading.

VI. CONCLUSION

This study introduced and proposed an a meta-adaptive multi-horizon forecasting architecture, MG-FiLMT, for stock price movement classification. In contrast to the traditional models relying exclusively on historical prices and also on simplified technical indicators, the proposed architecture uses market structure cues such as FVG, OB characteristics, and rolling-window hurst exponents, so that the model can be programmed to adapt to liquidity fluctuations, volatility changes, and institutional trading footprints. This structured representation is then processed via a coordinated set of modules using a variable selection network for dynamic feature gating, a multi-resolution projection block that captures multiple scale temporal dependencies, and a FiLM-based conditioning layer that allows for horizon-specific adaptation both in the encoder and decoder. The model also extends predictions to 3, 7, and 10-day horizons to support multi-horizon forecasting beyond the standard 1-day ahead.

A comprehensive and extensive experiments were carried out for eight indices using the time series cross-validation (3, 5, and 7-fold) to ensure no look-ahead bias of the model. The proposed model outperformed all considered benchmarks, and the ablation studies illustrated the contribution of the model's components both individually and together. A paired t-test confirmed that the proposed MG-FiLMT yields statistically significant improvements over the compared variants and few state-of-the-art benchmarks. Overall, this study highlights and emphasizes innovative ways of

improving deep learning models for stock market prediction with explicit structural market information, combined with adaptive multi-horizon modeling, that brings significant gains classification wise and profitability. The proposed MG-FiLMT forms a solid and extendable basis for real-world trading systems and provides a well-balanced approach that serves both academic research and practical applications of algorithmic trading.

DATA AVAILABILITY

All data used in this study will be made available by the first author upon request.

CONFLICT OF INTEREST

The authors declare no conflict of interest as defined by IEEE, or other interests that might be perceived to influence the results and/or discussion report in this paper.

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